
Topic 03: Finding Axes of Maximum Truth: Eigen & SVD

Applied Linear Algebra via NumPy

TOPIC 03

Finding Axes of Maximum Truth: Eigen & SVD

Invariant directions under transformation, spectral theorem, and PCA from scratch.

Cognitive Objectives

By the end of this topic, learners will be able to:

- **Identify invariant directions** geometrically where linear transformations act as pure scalars ($\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$).
- **Diagonalize symmetric operators** using the Spectral Theorem on data covariance matrices $\Sigma = \frac{1}{n}\mathbf{X}^T\mathbf{X}$.
- **Deconstruct SVD geometrically** as a 3-step pipeline: Rotation (V^T) $\rightarrow$ Scaling (Σ) $\rightarrow$ Rotation (U).
- **Implement PCA from scratch** using Singular Value Decomposition to compress high-dimensional feature spaces.

The High-Dimensional Redundancy Dilemma

The 50-Feature Multicollinearity Curse

- Customer transaction dataset contains $d = 50$ numeric columns.
- Many features (e.g., monthly spend, annual income, credit limit) are heavily correlated.
- **Curse of Dimensionality:** Models overfit, distance metrics degrade, latency escalates.
- **Challenge:** How do we extract the top 3 orthogonal directions in space that preserve 95% of all data variance?

The Discovery: Invariant Axes of Variance

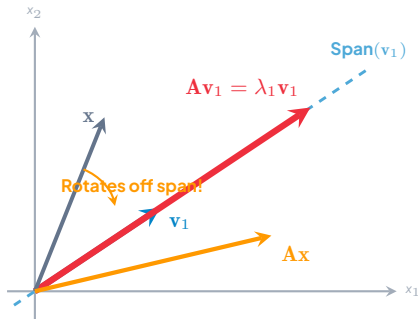
- Under transformation, most vectors rotate out of their original line of action.
- **Eigenvectors** never rotate; they only stretch by factor λ :

$$A\mathbf{v} = \lambda\mathbf{v}$$

- On the data covariance matrix, the top eigenvector points along the line of **maximum spread / variance!**

PUNCHLINE Eigenvectors find the natural coordinate system of the data.

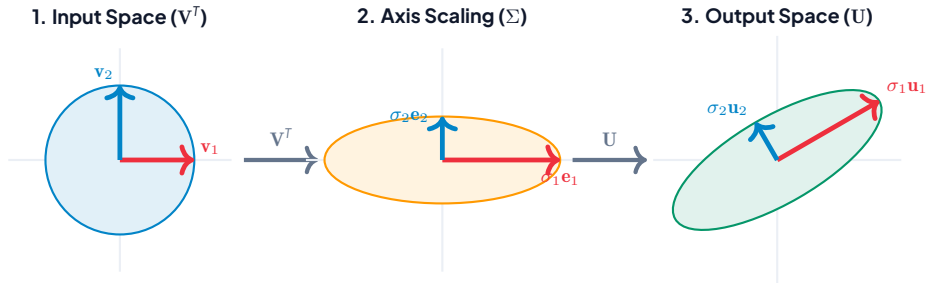
Hero Visualization: Eigenvectors as Invariant Directions



Invariant Directions Under A

- **General Vector x :** Multiplied by A , changes both length **and** direction ($Ax \not\parallel x$).
- **Eigenvector v :** Multiplied by A , remains on its original span line ($Av \parallel v$).
- **Eigenvalue λ :** The scalar stretch factor along the invariant line.
- **Spectral Theorem:** Any symmetric matrix $S = S^T$ has an orthonormal basis of eigenvectors ($S = Q\Lambda Q^T$).

Hero Visualization: SVD Geometric Decomposition



PUNCHLINE

Every real matrix $X = U\Sigma V^T$: Orthogonal Rotation $\rightarrow$ Axis Stretch $\rightarrow$ Orthogonal Rotation.

Production Checkpoint: Principal Component Analysis via SVD

PCA Mathematical Mechanics

For zero-centered data $\mathbf{X} \in \mathbb{R}^{N \times d}$:

1. Compute SVD: $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$.
2. Right singular vectors $\mathbf{V} \in \mathbb{R}^{d \times d}$ are the principal loading directions.
3. Project to k dimensions:
 $\mathbf{Z} = \mathbf{X}\mathbf{V}_{:,k} = \mathbf{U}_{:,k}\mathbf{\Sigma}_{:,k,k}$.
4. Explained variance ratio: $\frac{\sigma_i^2}{\sum \sigma_j^2}$.

NumPy Implementation from Scratch

```
1 import numpy as np
2
3 def pca_from_scratch(X: np.ndarray, k: int):
4     # 1. Center the features (zero mean)
5     X_centered = X - np.mean(X, axis=0)
6
7     # 2. Economy SVD decomposition
8     U, S, Vt = np.linalg.svd(X_centered, full_matrices
9                             =False)
10
11     # 3. Project data onto top-k components
12     Z = X_centered @ Vt[:k].T # shape: (N, k)
13
14     # 4. Compute explained variance ratio
15     explained_var = (S**2) / np.sum(S**2)
16     return Z, Vt[:k], explained_var[:k]
```